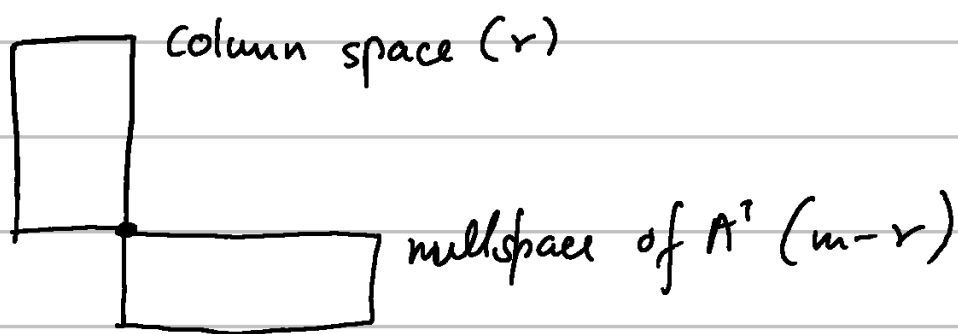
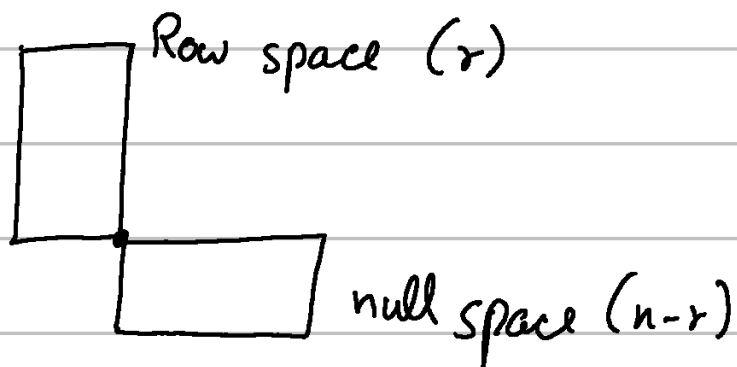
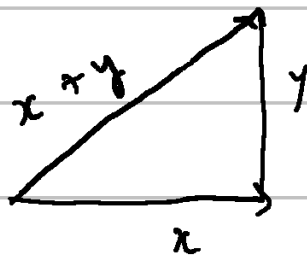


# Lecture 14



# These are some "naturally occurring" orthogonal subspaces.

⇒ Orthogonal vectors



$$x^T y = 0 \quad x, y, \dots \neq 0$$

Pythagoras

$$\|x\|^2 + \|y\|^2 = \|x+y\|^2$$

only when  $x^T y = 0$

$$x_1^2 + x_2^2 + \dots + x_n^2 + y_1^2 + \dots + y_n^2 = (x_1 + y_1)^2 + \dots$$

$$x_1^2 + y_1^2 + \dots = x_1^2 + y_1^2 + 2x_1 y_1 + \dots$$

$$\text{So, } 2(x_1 y_1 + x_2 y_2 + \dots) = 0$$

$$x_1 y_1 + x_2 y_2 + x_3 y_3 + \dots + x_n y_n = 0$$

$$\boxed{x^T y = 0}$$

Hence proved

# Dot product of 2 orthogonal vectors is zero.

# All vectors are orthogonal to the zero vector.

## Orthogonality of subspaces

⇒ Subspaces  $S$  and  $T$  are orthogonal  
if  $\forall v \in S$  &  $\forall u \in T$   
 $\boxed{v^T u = 0}$  i.e. orthogonal

⇒  $S$  &  $T$  are orthogonal then  
their  $S \cap T$  is only  $\{0\}$

# Row space is orthogonal  
to the null space.

$$\Rightarrow x_n \in N(A)$$

$$x_n = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$A x_n = 0$$

$$\begin{bmatrix} A_1 x_n \\ A_2 x_n \\ \vdots \\ A_m x_n \end{bmatrix} = 0 \quad \text{where } A_i \text{ is the } i^{\text{th}} \text{ row of } A \text{ of dim } 1 \times n$$

So, all rows are orthogonal to every  $x_n$

$$S = c_1 A_1 + \dots + c_m A_m$$

$$T = d_1 x_1 + \dots + d_t x_t$$

$$S \cdot T = d_1 x_1 \overbrace{(c_1 A_1 + \dots + c_m A_m)}^0 + d_2 x_2 \overbrace{(\dots)}^0$$

$$\boxed{S \cdot T = 0}$$

# And by a very similar proof, we can prove the column space is orthogonal to the nullspace of  $A^T$ .

eg  $A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix}$

$$n = 3 \quad \text{rank} = 1$$

$$\dim(\text{row space}) = 1$$

$$\dim(N(A)) = 2$$

$$\text{row space} = \lambda \begin{bmatrix} 1 & 2 & 5 \end{bmatrix}$$

$$\text{null space} = c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix}$$

# Null space & row space are orthogonal complements i.e. their dimensions add up to the whole space.

$\Rightarrow$  Null space contains all vectors that are  $\perp$  to row space.

$\Rightarrow Ax = b$ , "Solve" when there is no solution. As close as possible  
 $m > n$

- Used when there are a lot of variables/readings & we want to separate out "junk" or noise.

Key matrix  $\rightarrow$   $\boxed{A^T A \hat{x} = A^T b}$   $\hat{x} \rightarrow$  best solution  
when  $Ax = b$  is unsolvable

$$\underline{\underline{A^T A}}$$

→ square  $n \times n$  matrix

→ symmetric

$$\rightarrow N(A^T A) = N(A)$$

$$\rightarrow \text{rank}(A^T A) = \text{rank}(A)$$

eg  $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 + 2 + 5 \\ 8 & 1 + 4 + 25 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 8 & 30 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}$$

$A^T A \rightarrow \text{non-invertible}$

invertible

#  $A^T A$  is invertible when  $N(A^T A)$  is only zero vector i.e.,  
 $\text{rank}(A^T A) = n$

• Since  $\text{rank}(A^T A) = \text{rank}(A)$ ,

#  $A^T A$  is invertible when  $\text{rank}(A) = n$